

OBENG FRANCIS

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EDUCATION

Virginia Tech

Blacksburg, VA, USA

Ph.D. (In Progress), Computer Science GPA 4.0/4.0

Expected 2027

M.S. Computer Science GPA 4.0/4.0

December 2025

Graduate Certificate: Future Professoriate

December 2025

Advisor: [Dr. Eli Tilevich](#)

University of Education

Winneba, Ghana

B.S. Info. Technology Edu. (First Class, CGPA 3.67/4.0)

July 2019

Enchi College of Education

Enchi, Ghana

3-year Diploma in Basic Education

July 2012

AWARDS AND HONORS

- **Pratt Fellowship Award**, Department of Computer Science, Virginia Tech 2026–27
- **CIRTL at Virginia Tech Associate Teaching Badge:**
Center for Excellence in Teaching and Learning, Virginia Tech March 2026

RESEARCH INTEREST

Software engineering (fault tolerance—reliability and safety, AI-integrated software reliability and safety, automation), edge computing, mobile computing, and Digital Education.

RESEARCH EXPERIENCE & PUBLICATIONS

Virginia Tech—Software Innovations Lab

Blacksburg, VA, USA

Graduate Research Assistant

August 2023—Present

DeepGuard: Adaptive, Cost-Aware Retry Control for Cloud-Native Microservices via RL
Under Review—Middleware 2026

- Designed a PPO-based retry controller using Stable-Baselines3, and Gymnasium with a 6-feature MDP: reward balances service success rate against operational cost, [🔗 Code](#).
- Evaluated on Alibaba production traces and 3 synthetic overload scenarios; achieved CER gains of +1.15 to +2.68 over the best Threshold baseline across five of six conditions.

Fault-Handling Practices in Open-Source Web Service Client Repositories Across Languages
Under Review—ICSOC 2026

- Built a Python-based static analysis framework to identify and categorize exception strategies across 9 languages in 16K+ GitHub repositories. [🔗 Code](#),
- Trained an XGBoost classifier on 4,246 labelled repos, yielding Macro- $F_1 = 0.62 \pm 0.015$ and Cohen's $\kappa = 0.58$ against human labels. [🔗 Code](#).

Toward a Taxonomy of Emerging Failure Modes in Software Systems. (*Under Review—ESEM 2026*)

- Created a taxonomy of emerging failure modes in modern software systems using 100 real-world failure incidents. [🔗 Code](#).

Architecting Certainty: A New Framework for AI-Driven Software Reliability. (*Manuscript in preparation*)

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C++, PHP, HTML/CSS, SQL
- **DBMS:** SQL, MySQL, SQLite
- **Research/Data Analysis Tools:** LaTeX (Overleaf), Tableau, JMP, Excel, and SPSS
- **Framework and Libraries:** PyTorch, NumPy, Pandas, Matplotlib, Vue.js, Bootstrap, XGBoost, JMeter
- **IDEs/VC/Cloud:** VS Code, IntelliJ IDEA, Jupyter, PyCharm; Git, GitHub, Docker; Google Cloud, Kaggle.

TEACHING EXPERIENCE

Virginia Tech

Blacksburg, VA, USA

Graduate Teaching Assistant

2023 – Present

Courses Taught:

- | | |
|--|-------------------|
| • Data Structures and Algorithms (Undergraduate) — Java | 50 Students |
| • Mobile Application Development (Graduate) — Kotlin | 45 Students |
| • Web Application Development (Graduate) — HTML, CSS, JavaScript | 76 Students |
| • Software Engineering (Graduate) — Java | 35 Students |
| • Introduction to Software Design (Undergraduate) — 2 Semesters | 94 Students total |
| • Professionalism in Computing (Undergraduate) — 2 Semesters | 92 Students total |

Key Responsibilities

- Leading programming lab sessions twice per week for the entire semester.
- Holding office hours to assist students in overcoming their challenges in the course project.
- Reviewing students' applications (software) development process to ensure it aligns with project requirements (App Debugging and Reviews).
- Grading students' assignments and projects for the assigned course.
- Reporting on students' graded assignments and projects.

Obuasi Senior High Technical School

Obuasi, Ashanti – Ghana

Information Technology Instructor

Oct. 2019 – Dec. 2022

- Instructed 150+ high school students in C++, QBasic, HTML, CSS, Access, and SQL.
- Improved student programming proficiency by 95% through hands-on project-based learning.
- Developed a comprehensive curriculum resulting in a 100% student pass rate in the national ICT examinations for Senior High Students.
- Created two instructional textbooks, adopted by the ICT department at Obuasi Senior Technical School.

West Africa Examination Council

Obuasi, Ashanti – Ghana

Assistant Examiner

Nov. 2020 – Dec. 2022

- Evaluated 1,000+ ICT examination scripts annually.
- Maintained 99.5% accuracy in grading with quality assurance protocols.
- Contributed to the examination standards development for the national ICT curriculum.

Ghana Education Service

Ashanti Region – Ghana

Professional Info. Comm. Tech. Instructor/Teacher

Sept. 2012 – Aug. 2015

Impact:

- Taught computer literacy to 200+ junior high school students (ages 11–15).
- Established computer lab protocols, increasing student computer access by 85%.
- Achieved 90% improvement in student digital literacy assessment scores.

INSTRUCTIONAL MATERIALS AVAILABLE ON AMAZON

Obeng, F. (2022). *Information and Communications Technology for Senior High School Students*. Developed and compiled two comprehensive instructional books for high school ICT education.

These two books were used as a classroom resource during my time at Obuasi Senior High Technical School.

- Information and Communications Technology (**Elective Theory**)—200+ pages covering fundamental computing concepts, data representation, introduction to programming languages, and algorithms. [View Book](#)
- Information and Communications Technology (**Elective Practical**)—180+ pages focused on hands-on programming exercises in QBasic programming, HTML, CSS, and access database management. [View Book](#)

PROFESSIONAL EXPERIENCE

Ghana Statistical Service

Asokwa, Ashanti – Ghana

District IT Officer

June 2021 – Nov. 2021

- Trained 256 supervisors and enumerators on census technological tools (tablets, laptops, Computer-Assisted Personal Interviewing—CAPI software).
- Supervised data transfer/syncing operations, ensuring 100% data integrity from the field to the national data warehouse.
- Implemented quality control protocols, reducing data collection errors by 90%.

Data Collection Specialist (Interviewer)

2017 – 2018

- Conducted 1,000+ interviews for the census of agriculture using Computer-Assisted Personal Interviewing (CAPI).
- Managed maternal mortality survey data collection covering 1,000+ respondents.
- Maintained 98% data quality standards throughout the collection period.

RESEARCH COMPLIANCE

- Financial Conflict of Interest Training Requirement for Sponsored Research.
- Information Privacy and Security (IPS).
- Basic Responsible Conduct of Research.

PROJECTS AND TECHNICAL ACHIEVEMENTS

BookStore E-commerce Platform

Web Application Development Course

- Developed a full-stack e-commerce platform using HTML, CSS, JavaScript, Vue.js, Java, RESTful APIs, and Data Access Object (DAO) pattern.
- Implemented secure payment gateway integration and user authentication system.
- Achieved 100% functionality requirements with responsive design supporting mobile devices.

Guest House Reservation System

Internship

- Created a web-based reservation management system with real-time availability tracking.
- Integrated SMS notification system for booking confirmations.
- Deployed using MySQL database with 99.9% uptime performance.

School Website and Exeat Application System Prototype

Academic Project

- Developed a comprehensive school management system with an integrated online exeat (exit permit) application portal.
- Created a responsive web interface supporting 1,000+ concurrent users during application periods.
- Implemented automated application processing, reducing manual review time by 80%.

- Integrated document upload system with real-time status tracking for applicants.

Online Bus Ticketing System

Undergraduate Capstone Project

- Designed and developed a comprehensive bus ticketing platform serving 1,000+ daily users.
- Implemented route optimization algorithms, reducing travel time calculations by 25%.
- Integrated mobile payment systems that support multiple payment methods.

CERTIFICATIONS AND ONLINE COURSES

University of Michigan

Ann Arbor, Michigan

- Programming for Everybody (Getting Started with Python)
- Web Application Technologies and Django

June – July 2021

June 2023

University of Illinois

Urbana-Champaign

- Data Visualization

June – July 2021

DeepLearning.AI

USA

- Artificial Intelligence for Everyone

May 2023

University of Pennsylvania

USA, Pennsylvania

- Introduction to Python Programming

July 2023

University of London

UK, England

- Introduction to Network Security

June 2023

Microsoft

- Intro. Computers and Operating Systems, and Security

Aug 2023